

PENGUIN ACCIDENT INVESTIGATION BOARD

PENGUIN'S DEMOCRATIC REPUBLIC OF PATAGONIA • OFFICE OF FLIGHT OPERATIONS RESEARCH

Crosswind Landing Downwind Drift Rate Investigation

Boeing 747-400 — Autopilot-Disengaged Ground-Track Response to Ambient Crosswind

Report Number	PAIB/2026/015
Investigation Type	Technical Performance Review
Aircraft Type	Boeing 747-400 (simulated, PSX)
Trigger	Pilot report of unrealistically fast downwind drift during crosswind de-crab
Date of Issue	15 August 2026
Status	Open — follow-up recommended (see Recommendations)
Investigator-in-Charge	Claude, Automated Test & Analysis Unit

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"In Claude Confidimus"

Executive Summary

This investigation was opened following a type-rated pilot's report that, during a crosswind landing, the aircraft "seems to start drifting downwind much quicker than in real life (or FFS for that matter)" once de-crabbing during the flare, making it difficult to fly the wings-level, de-crab-to-touchdown technique the pilot normally uses on the real aircraft and in a full-flight simulator. The Board built a passive, non-interfering monitoring harness — one that logs aircraft position, heading, bank, and ground-contact state at high rate but never writes any control input to the simulator — and used it to measure the aircraft's actual downwind ground-track drift rate across a matrix of pilot-commanded crosswind conditions (5, 10, 20, 30, and 40 kt direct crosswind, a 30 kt/45° quartering crosswind, and a calm-wind control), with the pilot manually flying each approach on autopilot/ILS to a stabilized centerline track, then disconnecting the autopilot and releasing motion freeze to expose the aircraft to the commanded wind with no further corrective input of any kind.

Across every condition tested, the aircraft's downwind ground-track drift rate converged to the commanded crosswind component to within approximately 2–4%, and did so quickly: within 1.6 seconds of motion release in the slowest case, and in several runs at or below the harness's own ≈0.2-second sampling resolution — i.e. no measurable ramp-up at all. The Board considers this consistent with ordinary wind-triangle physics for an aircraft with no active heading or bank correction, rather than evidence of a simulation defect in this specific (fully hands-off) configuration. However, this passive test methodology is not the same maneuver the reporting pilot described: it cannot by itself confirm or refute whether PSX responds correctly to *active* corrective control input during a real de-crab. That specific comparison is identified as necessary follow-up work and this investigation remains open pending it.

> **KEY FINDING**
> With autopilot disengaged and no corrective control input applied at any point, the aircraft's downwind drift rate reaches the full commanded crosswind component (5–40 kt tested, plus a 45° quartering case) within 1.6 seconds of motion release in every run, and in several runs within the harness's own sampling resolution. This is physically expected for a wings-level aircraft with no correcting input, and does not by itself establish whether the pilot's reported behaviour — occurring during an *active*, continuously-corrected de-crab — reflects a genuine simulator discrepancy. See Section 4 and Recommendation 1.

Scope of Trials

- Direct (90°) crosswind magnitude: 5, 10, 20, 30 (three repeats), and 40 kt
- Quartering crosswind: 30 kt at 45° off the runway heading (headwind + crosswind component)

- Calm-wind control run (0 kt commanded) to establish the harness's own measurement noise floor
- All runs: EHAM runway 06, aircraft established on autopilot/ILS on the extended centerline prior to release, motion freeze on until manually released by the pilot after autopilot disconnect
- Wind-direction input convention (true vs. magnetic heading reference) empirically verified against the aircraft's own IRS-driven navigation display wind indication

1 Background

A type-rated pilot reported: *"I find that during crosswind landing de-crabbing, the aircraft seems to start drifting downwind much quicker than in real life (or FFS for that matter), making it quite hard to do what I normally do in real life (de-crab during flare, wings level until touch down) under certain cross wind/runway conditions."* This describes a specific, active maneuver: crabbing into wind on final approach, then transitioning to a wings-level attitude during the flare while continuously applying small corrective inputs to hold the runway centerline, with the crosswind correction applied progressively rather than removed all at once.

To make the underlying aircraft response directly measurable and repeatable, the Board and the reporting pilot agreed on a simplified, controlled first test: establish the aircraft on autopilot/ILS under calm wind conditions, aligned with the runway centerline, with motion freeze engaged; then apply a step change in wind, disconnect the autopilot, and release motion freeze — producing an aircraft that is, from that instant, wings-level, on runway heading, with no autopilot and no pilot control input of any kind, exposed to the full commanded crosswind. This isolates the aircraft's bare aerodynamic and inertial response to crosswind from any question of pilot or autopilot control-loop behaviour, and was explicitly understood by both parties as a starting point rather than a full reproduction of the pilot's actual complaint (see Section 4 and Recommendation 1).

2 Methodology

The Board built a dedicated monitoring harness that connects to the simulation host over its native network protocol in a purely passive role: it subscribes to the aircraft's position/attitude telemetry, ground-contact state, height, and autopilot mode string, and logs every update at the host's native rate (≈5 Hz), but at no point sends any control input, weight change, wind change, or other write to the simulator. All test conditions (aircraft positioning, wind setting, autopilot disconnect, motion freeze release) were set up and executed manually by the reporting pilot; the Board's role was limited to instrumented observation.

1. **Reference heading capture** — on connecting for each run, the harness waits until the aircraft is airborne and holds a stable heading (re-arming this wait if the aircraft is found on the ground, e.g. from a prior run's rollout not yet reset), then averages ten

heading samples using a circular mean to establish a reference (runway) heading. This is captured while the aircraft is still under calm-wind autopilot/ILS control, tracking the centerline, so it does not depend on knowing the runway's published heading or its magnetic variation.

2. **Ground-track derivation** — instantaneous ground speed and track angle are computed from consecutive latitude/longitude samples via a local flat-earth (east/north) projection, rejecting sample pairs with negligible displacement (motion-freeze artefacts) or with a timing gap exceeding 1 second.
3. **Crosswind/along-track decomposition** — the ground velocity vector is resolved against the reference heading into a crosswind component (perpendicular to the runway) and an along-track component (parallel to it), which is the quantity of interest for downwind drift.
4. **Touchdown detection and auto-stop** — the harness detects the ground-contact transition and continues logging briefly afterward before stopping automatically, so each run captures the full airborne exposure to the commanded wind plus the immediate post-touchdown transition.
5. **Steady-state analysis** — per run, the point of motion release is identified from the logged height first departing from its pre-release value; the crosswind component is then averaged over all airborne samples from six seconds after that point to touchdown, excluding the initial transient.

2.1 Wind-Direction Input Convention

No variable on the simulator's network protocol reports ambient wind directly, so the actual crosswind angle achieved cannot be read back independently of the aircraft's own inertial response — a limitation directly relevant to this section. The first trial run (Section 3, Run 1) commanded a wind direction equal to the runway's displayed/AP-tracked heading plus 90°, intending an exact direct crosswind. The resulting steady-state drift measured slightly but consistently below the commanded 30 kt (29.3 kt, Section 3.1), suggesting the commanded wind was not quite perpendicular to the runway. The reporting pilot subsequently identified a plausible cause: the wind-direction input may be referenced to true heading, while the aircraft's displayed/AP-tracked heading is magnetic. This investigation did not independently verify which convention the wind input itself uses, so no root cause is asserted; instead, the pilot verified the intended geometry directly and empirically, using the aircraft's own IRS-driven navigation display wind readout, and found that a commanded wind direction of 150° (rather than the runway heading $+90 \approx 147.5°$ that a same-convention assumption would suggest) produced an on-instrument reading of an almost exact 90° crosswind. All subsequent 90°-crosswind runs (Section 3, Runs 2–9) used this pilot-verified 150° wind direction; the 45°-quartering run (Run 10) used 105°, the same offset applied consistently. This is flagged here as a practical methodology hazard for anyone setting up a similar test in PSX, not as a finding about the pilot's original complaint.

3 Findings

3.1 Steady-State Drift vs. Commanded Crosswind Magnitude

Nine runs were flown at direct (90°) crosswind, spanning 5–40 kt, plus one further run at 30 kt discarded for a methodological reason noted below. Steady-state crosswind drift is the mean of the crosswind component of ground track over all airborne samples from six seconds after motion release to touchdown.

Run	Commanded Crosswind	Measured Steady-State Drift	Std. Dev.	Deviation
9	5 kt	5.21 kt	0.11 kt	+4.1%
8	10 kt	10.22 kt	0.23 kt	+2.2%
7	20 kt	20.66 kt	0.58 kt	+3.3%
1	30 kt *(uncorrected wind heading, Section 2.1)*	29.33 kt	0.55 kt	−2.2%
2	30 kt	30.69 kt	0.93 kt	+2.3%
3	30 kt	30.11 kt	0.83 kt	+0.4%
5	30 kt	30.76 kt	0.72 kt	+2.5%
6	40 kt	40.70 kt	0.89 kt	+1.8%

Run 4 (a fourth 30 kt repeat) is excluded: the harness happened to start, and lock its reference heading, during ground rollout from the previous run before the aircraft had been repositioned for the next approach, producing an incorrect reference heading and an artificially inflated apparent drift figure in that run's raw log. The harness was corrected immediately afterward to wait for a stable airborne heading before capturing the reference, and the fault did not recur. The run's raw data is retained in the archive for transparency but is not used in this report's figures.

Runs 2, 3, and 5 (three independent 30 kt repeats under the pilot-verified wind heading, Section 2.1) average 30.52 kt, a tight spread (30.11–30.76 kt) consistent with run-to-run reproducibility rather than any systematic trend. Across the full magnitude sweep, measured drift runs consistently 1.8–4.1% above the commanded value (Run 1 being the exception, for the wind-heading reason already discussed). The Board considers this small, uniform, one-directional offset more likely attributable to residual imprecision in this investigation's own empirically-captured reference heading — itself subject to measurement noise of a similar order — than to a genuine simulator effect, and does not treat it as a finding in its own right.

3.2 45° Quartering Crosswind

Run 10 commanded a 30 kt wind at 45° off the runway heading (105°, per the offset established in Section 2.1), combining a crosswind and a headwind component. Geometrically, 30 kt at 47.25° off the reference heading (the wind bearing actually used, minus the measured reference heading) predicts a crosswind component of $30 \times \sin(47.25°) \approx 22.0$ kt.

Run	Commanded	Predicted Crosswind Component	Measured	Deviation
10	**30 kt @ 45°**	**22.03 kt**	**22.82 kt**	**+3.6%**

The measured value tracks the geometric prediction with the same small, uniform high bias observed throughout Section 3.1, and shows no sign of any additional effect specific to a quartering (as opposed to direct) crosswind.

3.3 Calm-Wind Control Run

Run 11 repeated the identical procedure (autopilot disconnect, motion freeze release) with no wind commanded, to establish the harness's own baseline measurement noise. A brief transient (peaking at +3.2 kt roughly one second after release) decayed to within ±0.7 kt by six seconds post-release and remained there for the balance of the approach.

Run	Commanded Crosswind	Measured Steady-State Drift	Std. Dev.
11	**0 kt (control)**	**+0.05 kt**	**1.19 kt**

This confirms the harness introduces no meaningful baseline drift or measurement bias of its own, and gives an approximate noise floor (≈1 kt standard deviation, briefly up to ≈3 kt during the immediate post-release transient) against which the Section 3.1/3.2 results can be judged.

3.4 Convergence Time to Steady State

For each run, the time from motion release to the crosswind component first entering, and remaining within, ±10% of that run's own steady-state mean was measured directly from the logged track.

Run	Commanded Crosswind	Time to ±10% of Steady State
9	5 kt	≤0.2 s *(within sampling resolution)*
8	10 kt	≤0.2 s *(within sampling resolution)*
7	20 kt	0.43 s
2	30 kt	1.31 s
3	30 kt	0.66 s
5	30 kt	1.49 s
6	40 kt	≤0.2 s *(within sampling resolution)*
10	**30 kt @ 45°**	**1.58 s**

In every run, the crosswind component reached steady state within 1.6 seconds of motion release, and in three of eight runs did so within the harness's own ≈0.2-second sample interval — i.e. no ramp-up was resolvable at all. No relationship between commanded crosswind magnitude and convergence time is apparent in this data.

3.5 Post-Touchdown Behaviour (Observed, Not Analyzed)

In every run, the measured crosswind component decayed over the several seconds following touchdown, alongside a corresponding heading change toward the wind, visible consistently across all eleven runs' raw data. This is consistent with weathervaning and/or directional ground control (rudder, nosewheel steering, differential braking) acting during rollout — a materially different physical regime from the pre-touchdown airborne drift this investigation was designed to measure. The Board did not analyze this behaviour further, as it falls outside the scope of the reporting pilot's original complaint (which concerns the pre-touchdown de-crab and flare), but notes it for completeness and as a possible topic for a separate investigation (Recommendation 3).

4 Discussion

Across the full tested range — 5 to 40 kt direct crosswind and a 30 kt/45° quartering case — the aircraft's downwind ground-track drift rate, measured with autopilot disengaged and no corrective control input of any kind, converges to the commanded crosswind component to within approximately 2–4%, and does so within 1.6 seconds of motion release in every case, in several runs at or below the resolution of the harness's own sampling. The Board considers this behaviour consistent with straightforward wind-triangle physics rather than evidence of a defect in this specific configuration: for an aircraft holding a fixed heading and bank with no active correction, ground velocity is simply the vector sum of the airspeed vector and the ambient wind vector, and there is no aerodynamic mechanism that would produce a gradual build-up of the crosswind component once that wind is present and nothing opposes it. What can limit how *quickly a pilot perceives* the resulting deviation is not airframe response time but how promptly the deviation becomes visible against the runway picture — a perceptual and technique question, not a physics one, and one this passive test cannot speak to.

This is precisely why this test methodology, deliberately chosen for its simplicity and repeatability, cannot by itself confirm or refute the reporting pilot's actual complaint. The pilot described an *active*, continuously-corrected de-crab: crabbing into wind on approach, then transitioning smoothly to wings-level during the flare while applying ongoing small corrective inputs to hold the centerline, with the crosswind correction removed progressively rather than all at once. This investigation instead measured what happens when *no* corrective input is applied at all, at any point — a limiting case useful as a bound and a control, but not a reproduction of the maneuver the pilot actually flies. It remains entirely possible that PSX's response to *active* rudder/aileron correction during a real de-crab diverges from real-world or full-flight-simulator behaviour in a way this test could not detect, for example in control authority, sideslip build-up, or how quickly a given corrective input arrests the drift once begun — none of which this investigation exercised.

The small (1.8–4.1%), uniformly one-directional excess of measured drift over the commanded value, present across every magnitude and the quartering case, is too small and too consistent in character to indicate an anomaly; the Board attributes it tentatively

to residual imprecision in its own empirically-captured reference heading, itself derived from the same order of positional measurement noise as the drift figures it is compared against, rather than to any behaviour of the simulator itself.

Separately, the true-versus-magnetic wind-direction input ambiguity identified in Section 2.1 is a genuine practical hazard independent of the pilot's original complaint: a tester assuming the wind-direction input shares the aircraft's displayed heading reference will not, in fact, produce an exact 90° crosswind by adding 90° to the runway heading, as Run 1 of this investigation demonstrates directly.

5 Conclusions

1. With autopilot disengaged and no corrective control input applied at any point, the aircraft's downwind ground-track drift rate converges to the commanded crosswind component to within approximately 2–4% across the full tested range (5–40 kt direct crosswind, and a 30 kt/45° quartering crosswind).
2. Convergence to steady-state drift is fast in every configuration tested: within 1.6 seconds of motion release in the slowest case, and at or below the harness's own ≈0.2-second sampling resolution in three of eight runs — i.e. no measurable ramp-up.
3. This behaviour is consistent with expected wind-triangle physics for a wings-level aircraft with no active correction, and is not, in this specific (fully hands-off) configuration, evidence of a simulation defect.
4. This investigation's passive, hands-off test methodology cannot confirm or refute the reporting pilot's original complaint, which specifically concerns behaviour during an *active*, continuously-corrected de-crab — a maneuver this investigation did not exercise.
5. A calm-wind control run showed effectively zero steady-state drift (+0.05 kt, well within measurement noise), confirming the harness introduces no baseline bias of its own.
6. PSX's wind-direction input does not appear to share the aircraft's displayed/AP-tracked heading reference frame (true versus magnetic); commanding wind at the runway heading ±90° did not, in this investigation's first run, produce an exact 90° crosswind as independently verified against the aircraft's own IRS-driven navigation display wind indication.

6 Recommendations

1. **Highest priority:** repeat this investigation using an actively-flown de-crab technique — a pilot crabbing into wind on approach, then transitioning to wings-level during the flare with continuous small corrective control inputs, exactly as described in the original report — with the same passive monitoring harness recording throughout. This is the specific maneuver the original complaint concerns and this investigation has not yet tested it.

2. Any future PSX wind-related test setup should independently verify the actual crosswind angle achieved via the aircraft's own navigation display wind indication, rather than assuming the wind-direction input shares the aircraft's heading reference frame (Section 2.1).
3. The post-touchdown crosswind decay/weathervaning behaviour noted in Section 3.5 was outside this investigation's scope but may merit a dedicated follow-up if further reports concern rollout directional control rather than pre-touchdown drift.

7 References

This investigation relied entirely on the Board's own instrumented monitoring harness and did not consult any external manufacturer or training-manual source; no figure in this report is drawn from an outside document. The pilot report quoted in Section 1 was relayed directly to the Board by the reporting pilot and is not a public forum citation.

8 Acknowledgements

The Board thanks the reporting pilot for bringing a precisely-worded, testable complaint, and for personally flying every approach, wind change, and autopilot disconnect in this investigation's entire eleven-run test matrix — including patiently re-flying one run after the harness's own reference- heading bug (Section 3.1) was caught and fixed mid-session. The Board also gratefully acknowledges **CONAIR Technics**, and in particular its Vice President, **Señor Cousin**, for wind-generation equipment support throughout this investigation's crosswind trials, and the **Fuerza Aérea de Liberación Pinguino (FALP)** for continued airfield access and general cooperation. Their support is reflected in the quality of the evidence this report is able to present.

— Chief Investigator Claude, PAIB

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